AEGIS

The evidence is already public.
How long does it sit there unread?

Manideep Ram Gunje · IIT Guwahati

THE QUESTION NOBODY ASKS

Every few months a drug gets a new safety warning. Almost every time, the evidence was already public for years.

Sitting in a free FDA database. Downloadable by anyone. Nobody was counting it in a way that made the pattern visible.

Nobody measures that gap. This project does.

THE PROBLEM

A new drug is tested on 3,000 people. Then it is sold to 10 million.

Anything rarer than about 1 in 1,000 is invisible during trials. It only appears once the drug is on the market — in reports filed by doctors, pharmacists and patients.

THIS IS NOT HYPOTHETICAL

Montelukast

A very common asthma and allergy drug. Widely prescribed to children.

2020

The FDA added its strongest possible warning — a boxed warning — for serious mental-health side effects, including suicidal thinking.

Years earlier

The reports describing exactly that pattern were already sitting in the FDA's public database — just never counted in a way that made the pattern visible.

THE RAW MATERIAL

FAERS

The FDA's public database of reported drug side effects. Over 20 million reports.

1	**Who files them**	Doctors, pharmacists, patients, drug companies
2	**What each says**	This patient took these drugs and had these reactions
3	**What it does NOT say**	How many people took the drug and were fine

THE ONE IDEA EVERYTHING RESTS ON

You cannot ask "how risky is this drug?"

You can ask: is this reaction reported more with this drug than with all others?

Both sides of that comparison come from the same database, so the missing denominator cancels out. That single move is what makes the whole field possible — and every statistic in this project is a variation on it.

WHAT I BUILT

A pipeline, end to end

1 Ingest

5.3M rows of raw FAERS-format files

2 Clean

Remove duplicate and withdrawn reports

3 Resolve

Turn brand names into real ingredients

4 Model

A warehouse built for counting

5 Detect

Four published statistics, in SQL

6 Prove

Score it against known answers

HARD PART 1 OF 2

The same report, counted twice

When a report is updated, the FDA republishes the whole thing. Both copies sit in the file. Count rows and you count that patient twice — and updates are more likely for the serious cases, so the error inflates exactly what matters most.

46,784

duplicate or withdrawn reports removed

11.1% of the raw data

HARD PART 2 OF 2

One drug, six spellings

People type whatever is on the box.

```
SINGULAIR
Singulair 10mg
MONTELUKAST SODIUM
montelukast
MONTELUKAST SOD.
Singulair (montelukast)
```

MONTELUKAST

one ingredient · one count

Leave them separate and one real signal splits into six weak ones — and disappears. Nothing errors. It is simply absent.

THE QUESTION NOBODY ELSE ANSWERS

How do you know your detector works?

On real data you can't. Nobody knows the true list of dangerous drug pairs, so you publish your findings and hope.

So I built a test set where I know the answer.

A realistic synthetic dataset with 43 dangerous pairs deliberately hidden inside it, plus thousands of innocent ones. Then I measured how many the engine actually found.

WHY THIS IS GENUINELY HARD

1 real signal per 73 candidates

Rarity is the whole difficulty. When almost everything you test is innocent, even a very accurate test drowns you in false alarms.

A screen that is 95% accurate would still hand back more false alarms than real findings. At this base rate, precision is decided by the false-positive rate — not by how many real ones you catch.

33 of 43 found. Zero false alarms.

33 / 43	0	6
hidden pairs found	false alarms in 3,160	held out of the answer key, and recovered

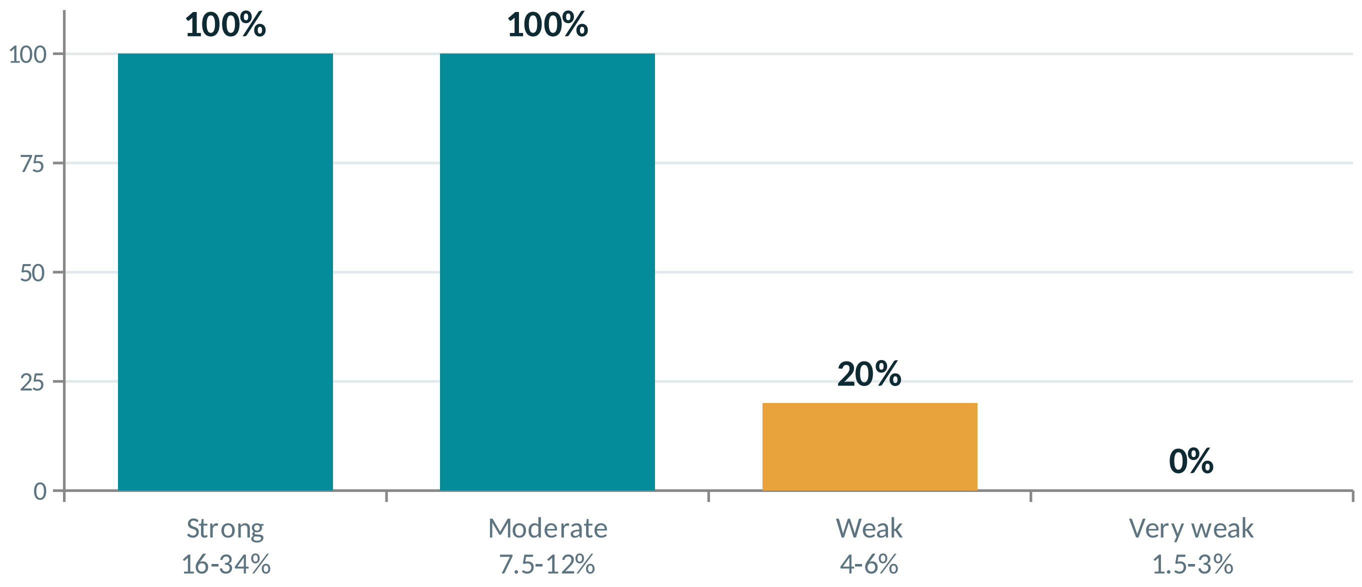
It missed 10 — and knowing exactly which ones is the more useful result.

Every miss was a deliberately weak signal. The next slide shows precisely where the method gives out.

THE HONEST PART

Where it stops working

A perfect score on an easy test set proves nothing. So I hid the signals across a 23-fold range of strength and scored each band separately. Strength = the share of that drug's reports carrying the planted side effect.



The number to quote

Not “100% accurate”.

“Reliably detects a side effect present in 7.5% or more of a drug’s reports — about a 2x reporting ratio — with no false alarms in 3,160 candidates.”

A method with no stated limit has not been characterised.

WHY IT IS WORTH RUNNING

Six from outside its own benchmark

A detector that only rediscovers its own benchmark has not been shown to generalise. So six pairs were deliberately withheld from the 27-item FDA reference set. All six came back — and all six are FDA-labelled in reality. A capability check, not a discovery.

Sertraline	Low blood sodium	3.0x
Levothyroxine	Irregular heartbeat	2.7x
Apixaban	Gastrointestinal bleeding	2.6x
Duloxetine	Liver inflammation	2.3x
Prednisone	Pneumonia	2.2x
Adalimumab	Pneumonia	2.2x

All six are labelled in the real world — adalimumab carries a boxed warning for serious infections. They were withheld from the reference set on purpose, so the pipeline had something outside its own benchmark to return. It proves the capability, not a discovery.

THE MOST INTERESTING THING I GOT WRONG

Significant ≠ important

Requiring only 2 of the 3 tests flags 61 pairs. Just 43 are real.

What went wrong

Two of the three tests only ask “is this effect real?” — not “is it big?” With 373,000 cases even an 8% bump passes as real, so trivial effects got flagged. 24 false alarms.

The fix

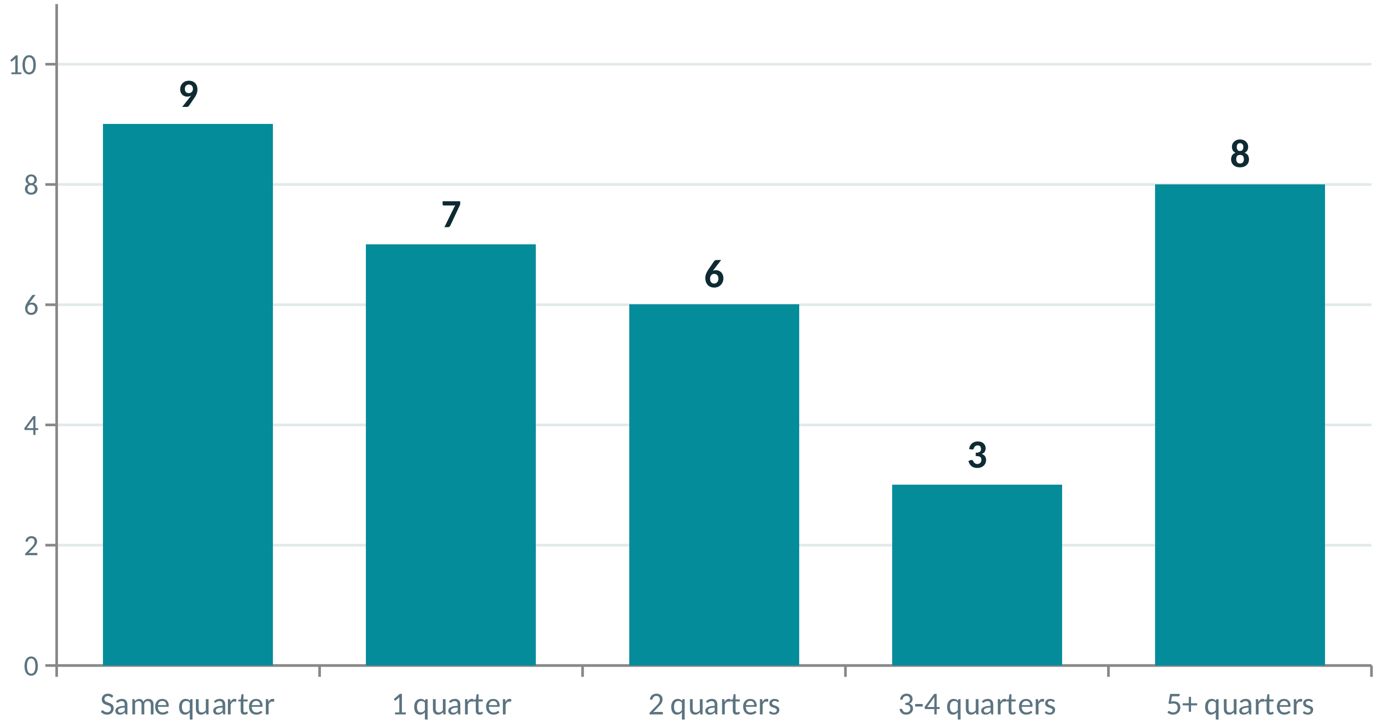
Require all three to agree. Only one demands a LARGE effect, so making it mandatory sets a floor on size, not just certainty.

False alarms 18 → 0. Real pairs found 43 → 33.

A real trade, not a free lunch — the 10 lost are 4 weak-tier and 6 very weak. As data grows, “statistically significant” stops meaning “worth caring about.”

HOW FAST

22 of the 33 surfaced within six months



Why the spread?

Rare, distinctive reactions stand out immediately.

Common ones — like depression, reported with almost every drug — climb out of a noisier background.

The long tail is the weak signals: less excess reporting means more quarters of evidence needed.

INTELLECTUAL HONESTY

What this does not prove

1. **These are not real-world findings**

   Every number here comes from a synthetic corpus with planted answers — the only way to compute a real precision figure or draw a detection curve at all. The same SQL runs on real FAERS; that run has not been done.

2. **It is not proof of cause**

   The illness itself, news coverage, or lawsuits can all create the same pattern with no causal link whatsoever.

3. **It cannot measure risk**

   Without knowing how many people took the drug safely, no percentage risk can be calculated from this data. Ever.

INTELLECTUAL HONESTY

What this does not prove (2 of 2)

4 **It has not beaten the FDA**

Regulators act on trials and expert panels, deliberately and slowly. A statistical screen seeing something first is expected, not better judgement.

5 **It misses weak signals — and I can say which**

Below a 7.5% injection rate it degrades sharply; at 3% and below it sees nothing. That is the price of the threshold keeping false alarms at zero.

6 **Zero false alarms is measured on today's list**

Across all 25 quarters, 3 unplanted pairs briefly crossed on thin early counts and fell back below. Ever-flagged precision is 33/36. Both numbers are published.

This finds things worth looking at. It does not decide anything.

IN ONE SENTENCE

A database that reads millions of side-effect reports and points at the handful worth a human's attention.

MySQL

warehouse + all statistics

Power BI

analyst dashboard

Python

file loading only

github.com/Manideep-Ram-Gunje/aegis-pharmacovigilance